

Zixiang Wang

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Research Interest

I am a master's student at Peking University, advised by Prof. Liantao Ma. My research focuses on AI for Healthcare, especially medical large language models, electronic health record modeling, multi-agent collaboration, and reusable benchmark/toolkit infrastructure for clinical data analysis.

- (1) **Medical Large Language Models:** Building and evaluating LLM-driven systems for clinical prediction, reasoning, retrieval, and decision support.
- (2) **Electronic Health Record Modeling:** Modeling longitudinal, sparse, and multimodal EHR data with attention to clinical reliability and reuse.
- (3) **Benchmarks, Toolkits, and Protocols:** Creating open resources that make healthcare AI evaluation more reproducible and practically useful.

Education

Peking University

2024.09 - Present

School of Software and Microelectronics · M.Eng. in Software Engineering (Working on AI for Healthcare, medical large language models, and EHR modeling under the supervision of Prof. Liantao Ma.)

Beihang University

2019.09 - 2024.06

School of Software · B.Eng. in Software Engineering (Completed systematic training in software engineering and gradually shifted toward data modeling, machine learning, and healthcare AI research.)

Research Experience

Healthcare AI Publications

2024 - 2026

WWW / CIKM / STAR Protocols / Patterns / npj Digital Medicine / NeurIPS / CHI / KDD (Recent work spans methods, protocols, benchmarks, and systems for healthcare AI.)

Research Focus Formation

2023 - Present

Medical LLMs, clinical prediction, EHR modeling, benchmarks, protocols, and agentic healthcare systems (Research themes converged toward clinically grounded AI systems and reproducible healthcare data analysis infrastructure.)

Publications (* indicates equal contribution; † indicates corresponding author)

LLM Agents for Healthcare

- [WWW 2025] **ColaCare: Enhancing Electronic Health Record Modeling through Large Language Model-Driven Multi-Agent Collaboration**
Zixiang Wang*, Yinghao Zhu*, Huiya Zhao*, Xiaochen Zheng, Tianlong Wang, Wen Tang, Yasha Wang†, Chengwei Pan, Ewen Harrison, Junyi Gao†, Liantao Ma†
ACM International World Wide Web Conference (WWW), 2025
- [Preprint 2025] **HealthFlow: A Self-Evolving AI Agent with Meta Planning for Autonomous Healthcare Research**
Yinghao Zhu*, Yifan Qi*, Zixiang Wang, Lei Gu, Dehao Sui, Haoran Hu, Xichen Zhang, Ziyi He, Junjun He, Liantao Ma†, Lequan Yu†
Preprint, 2025
- [Preprint 2025] **ConfAgents: A Conformal-Guided Multi-Agent Framework for Cost-Efficient Medical Diagnosis**
Huiya Zhao*, Yinghao Zhu*, Zixiang Wang*, Yasha Wang, Junyi Gao, Liantao Ma†
Preprint, 2025
- [KDD 2024] **EHRFlow: A Large Language Model-Driven Iterative Multi-Agent Electronic Health Record Data Analysis Workflow**
Hao Wu*, Yinghao Zhu*, Zixiang Wang, Xiaochen Zheng, Ling Wang, Wen Tang, Yasha Wang, Chengwei Pan, Ewen Harrison, Junyi Gao†, Liantao Ma†
Artificial Intelligence and Data Science for Healthcare: Bridging Data-Centric AI and People-Centric Healthcare (KDD 2024 AIDSH Workshop), Oral, 2024

LLM for Healthcare

- [CIKM 2024] **EMERGE: Enhancing Multimodal Electronic Health Records Predictive Modeling with Retrieval-Augmented Generation**
Yinghao Zhu*, Changyu Ren*, Zixiang Wang, Xiaochen Zheng, Shiyun Xie, Junlan Feng, Xi Zhu, Zhoujun Li, Liantao Ma, Chengwei Pan†
ACM International Conference on Information and Knowledge Management (CIKM), 2024
- [Preprint 2024] **Prompting Large Language Models for Zero-Shot Clinical Prediction with Structured Longitudinal Electronic Health Record Data**
Yinghao Zhu*, Zixiang Wang*, Junyi Gao, Yuning Tong, Jingkun An, Weibin Liao, Ewen Harrison, Liantao Ma†, Chengwei Pan†
Preprint, 2024

Healthcare Benchmark

[npj Digital Medicine 2026] **ClinicRealm: Re-evaluating Large Language Models with Conventional Machine Learning for Non-Generative Clinical Prediction Tasks**
Yinghao Zhu*, Junyi Gao*, **Zixiang Wang***, Weibin Liao*, Xiaochen Zheng, Lifang Liang, Miguel O. Bernabeu, Yasha Wang, Lequan Yu, Chengwei Pan†, Ewen Harrison†, Liantao Ma†
npj Digital Medicine, 2026

[Preprint 2025] **MedAgentAudit: Diagnosing and Quantifying Collaborative Failure Modes in Medical Multi-Agent Systems**
Lei Gu*, Yinghao Zhu*, Haoran Sang*, **Zixiang Wang***, Dehao Sui, Wen Tang, Ewen Harrison, Junyi Gao, Lequan Yu†, Liantao Ma†
Preprint, 2025

[NeurIPS 2025] **MedAgentBoard: Benchmarking Multi-Agent Collaboration with Conventional Methods for Diverse Medical Tasks**
Yinghao Zhu*, Ziyi He*, Haoran Hu*, Xiaochen Zheng*, Xichen Zhang, **Zixiang Wang**, Junyi Gao, Liantao Ma†, Lequan Yu†
Conference on Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track, 2025

[Cell Patterns 2024] **A Comprehensive Benchmark For COVID-19 Predictive Modeling Using Electronic Health Records in Intensive Care**
Junyi Gao*, Yinghao Zhu*, Wenqing Wang*, **Zixiang Wang**, Guiying Dong, Wen Tang, Hao Wang, Yasha Wang, Ewen Harrison, Liantao Ma†
Cell Patterns, 2024

Healthcare Modeling

[KDD 2025] **Learnable Prompt as Pseudo-Imputation: Rethinking the Necessity of Traditional EHR Data Imputation in Downstream Clinical Prediction**
Weibin Liao, Yinghao Zhu, Zhongji Zhang, Yuhang Wang, **Zixiang Wang**, Xu Chu, Yasha Wang, Liantao Ma†
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), 2025

[CIKM 2024] **PRISM: Mitigating EHR Data Sparsity via Learning from Missing Feature Calibrated Prototype Patient Representations**
Yinghao Zhu, **Zixiang Wang**, Long He, Shiyun Xie, Xiaochen Zheng, Liantao Ma†, Chengwei Pan†
ACM International Conference on Information and Knowledge Management (CIKM), 2024

Trustworthy AI

[KDD 2024] **RetCare: Towards Interpretable Clinical Decision Making through LLM-Driven Medical Knowledge Retrieval**
Zixiang Wang*, Yinghao Zhu*, Junyi Gao, Xiaochen Zheng, Yuhui Zeng, Yifan He, Bowen Jiang, Wen Tang, Ewen Harrison, Chengwei Pan, Liantao Ma†, Ling Wang†
Artificial Intelligence and Data Science for Healthcare: Bridging Data-Centric AI and People-Centric Healthcare (KDD 2024 AIDSH Workshop), 2024

Toolkits & Platforms

[CHI 2026] **Augmenting Clinical Decision-Making with an Interactive and Interpretable AI Copilot: A Real-World User Study with Clinicians in Nephrology and Obstetrics**
Yinghao Zhu*, Dehao Sui*, **Zixiang Wang***, Xuning Hu, Lei Gu, Yifan Qi, Tianchen Wu, Ling Wang, Yuan Wei, Wen Tang, Zhihan Cui, Yasha Wang, Lequan Yu, Ewen Harrison, Junyi Gao, Liantao Ma†
ACM CHI Conference on Human Factors in Computing Systems (CHI), 2026

[KDD 2026] **OneEHR: Reproducible and AI Agent-Ready Longitudinal EHR Analysis Toolkit**
Yinghao Zhu, **Zixiang Wang**, Lei Gu, Dehao Sui, Yasha Wang, Ewen Harrison, Tianfan Fu, Junyi Gao, Lequan Yu†, Liantao Ma†
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD-26) Hands-On Tutorials, 2026

[STAR Protocols 2025] **Protocol for Processing Multivariate Time-series Electronic Health Records of COVID-19 Patients**
Zixiang Wang*, Yinghao Zhu*, Dehao Sui, Tianlong Wang, Yuntao Zhang, Yasha Wang, Chengwei Pan, Junyi Gao†, Liantao Ma†, Ling Wang†, Xiaoyun Zhang†
STAR Protocols, 2025

[SAIL 2025] **AIcare: An AI-Clinician Interaction System for Transparent and Actionable Clinical Decision Support**
Junyi Gao, **Zixiang Wang**, Dehao Sui, Wen Tang, Yasha Wang, Ling Wang, Liantao Ma, Yinghao Zhu
The Symposium on Artificial Intelligence in Learning Health Systems (SAIL), *Best abstract nominee*, 2025

[STAR Protocols 2024] **Protocol to process follow-up electronic medical records of peritoneal dialysis patients to train AI models**
Tianlong Wang*, Yinghao Zhu*, **Zixiang Wang**, Wen Tang†, Xinju Zhao, Tao Wang, Yasha Wang, Junyi Gao†, Liantao Ma†, Ling Wang†
STAR Protocols, 2024

Selected Project

Selected Honors and Awards

- ▷ **Outstanding Graduate of Beihang University**, Beihang University 2024
- ▷ **National Encouragement Scholarship**, Beihang University 2022, 2023

Services

- ▷ **Teaching Assistant**, Data Structures and Programming, Discrete Mathematics, Computer Hardware Fundamentals 2022 - 2023
- ▷ **Reviewer**, AAAI 2026 2025